Vadodara Institute of Engineering, Kotambi

Subject name: Object Oriented Programming-Java

Subject Code: DI04000021 **Discipline:** CE **Semester:** 4^{th}

Academic Year:2025 – 2026

Software: Notepad++

Sr No	Practicals
1	Create simple programs in Java. a. Create a Java program to print "Hello World", then compile, debug and execute it using Java compiler and interpreter. b. Create a Java program to perform arithmetic operations and demonstrate type casting.
2	Create Java programs using control structures. a. Create a Java program to find maximum of three numbers. b. Create a Java program to find sum of digits of a number
3	Create Java programs using arrays. a. Create a Java program to find sum and average of array elements. b. Create a Java program to perform addition of two matrices.
4	Create a Student class with fields and methods in Java program. Demonstrate object creation and method calling.
5	Create Java programs using constructor, this and static keywords. a. Create a Java program to demonstrate default and parameterized constructors and the use of 'this' keyword. b. Create a Java program to demonstrate different uses of 'static' keyword.
6	Create a Java program to demonstrate method overloading.
7	Create a Java program to perform various String operations of using built-in String methods.
8	Create a Java program to accept input using the Scanner class, store values and demonstrate autoboxing and unboxing
9	Create a Java program to demonstrate single and multilevel inheritance. (e.g., class Person → Employee → Manager)
10	Write a Java practical to declare a 1D array of 10 integers, accept values from the user, and find the maximum, minimum, sum, and average of all elements.
11	Write a Java practical to create a Student class with fields name, roll number, and marks using a parameterized constructor and a display() method to print details of multiple student objects.
12	Write a Java practical to demonstrate inheritance and method overriding by creating a parent class Animal and a child class Dog that overrides the sound() method using the super keyword.
13	Write a Java practical to demonstrate the Collections Framework by creating an ArrayList of student names to perform add, remove, and search operations and a HashSet to store and display unique roll numbers using a for-each loop.
14	Write a Java practical to calculate the percentage of a student based on marks of 5 subjects and display the corresponding grade using if-else ladder control structure.